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# CAMPUS GUIDE PROJECT PLAN

## 1.Introduction

The university campus environment is often expansive, complex, and dynamic. For incoming first-year students and visitors, navigating this environment can be a significant challenge. Currently, navigation relies heavily on static signage, paper maps, or asking for directions, all of which are inefficient and often lead to confusion. This project aims to bridge this gap by developing "CampusGuide," a mobile application designed to simplify campus navigation.

## 2.Problem statement

Students at University of Eswatini, particularly new and transfer students, face significant challenges navigating the campus environment. The lack of a dedicated digital navigation tool makes it difficult to locate buildings, classrooms, facilities, and services, resulting in late arrivals, frustration, and a poor overall campus experience.
Currently, students rely on printed maps, word of mouth, or generic mapping applications that are not tailored to the specific layout and needs of campus life. These alternatives are often inaccurate, outdated, or too general to be of practical use, especially for students who are new to the institution.
This project aims to develop a dedicated mobile application that provides interactive campus maps, turn-by-turn directions, and real-time campus information. The solution will empower students to navigate campus efficiently and confidently, improving their overall academic experience from day

## 3.Project Objectives

The core objectives of this initiative are to:

- Facilitate Location Finding: Assist users in identifying buildings, labs, classrooms, cafeterias, and amenities.
- Ensure Navigation: Provide accurate, turn-by-turn directions within the campus.
- Centralize Information: Showcase campus events, schedules, and urgent announcements.
- Enhance Accessibility: Deliver an intuitive, inclusive interface, including dedicated routes for students with disabilities.
- Improve Reliability: Enable offline functionality for areas with limited connectivity.

## 4. Target Users

The application is designed to serve the following users groups:

| User Group | Description | Primary Need |
|---|---|---|
| New / Freshman Students | Students in their first semester unfamiliar with campus | Full campus orientation & navigation |
| Transfer Students | Students joining from other institutions | Quick onboarding & layout understanding |
| Visitors & Guests | Parents, guests, or event attendees | Locating key venues and parking |
| All Students | General student population | Daily navigation and campus updates |

**5. Core Feature and Requirements**

The following features have been identified and prioritized based on user needs and project scope:

| Feature | Description | Priority |
|---|---|---|
| Interactive Campus Map | Visual, zoomable map of the entire campus | High |
| Search for Buildings/Rooms | Search bar to find any location on campus | High |
| Turn-by-Turn Directions | Step-by-step navigation from point A to B | High |
| Points of Interest | Library, clinic, cafeteria, ATMs, etc. | High |
| Campus Events & Announcements | Real-time updates and event listings | Medium |

| Feature | Description | Priority |
|---|---|---|
| Accessibility Routes | Ramps, elevators, and accessible pathways | Medium |
| Offline Mode | Core features available without internet | Medium |
| Emergency Contacts & Safety Info | Campus security, health services, etc. | Medium |
| AR Navigation | Camera overlay for real-time direction overlay | Low |

**5.Feasibility Analysis**

- Technical: Implementation is viable through standard cross-platform frameworks and existing map APIs (e.g., Google Maps, OpenStreetMap).
- Financial: Costs are expected to remain low-to-moderate by prioritizing open-source tools and free-tier APIs, with primary expenses related to labor and scaling.
- Time: An MVP is attainable within 3 to 6 months via a phased deployment strategy.
- Team: The team requires a frontend developer, backend developer, UI/UX designer, and project manager, with collective responsibility for QA.

**6. Risks and Mitigations**

| Risk | Likelihood | Impact | Mitigation Strategy |
|---|---|---|---|
| Inaccurate campus map data | Medium | High | Partner with administration for official floor plans and maps |
| Low student adoption | Medium | High | Involve students early via surveys and beta testing |
| API cost overruns | Low | Medium | Use free-tier APIs; set usage caps and monitor usage |

| Risk | Likelihood | Impact | Mitigation Strategy |
|---|---|---|---|
| Team capacity issues | Medium | Medium | Define clear roles; use Agile sprints with regular reviews |
| Scope creep | High | Medium | Lock MVP features early; defer low-priority features to v2 |
| Data privacy concerns | Low | High | Avoid storing personal data; comply with university privacy policies |

**7.Recommended Tech Stack**

| | Technology | Reason |
|---|---|---|
| Mobile App | Flutter or React Native | Cross-platform (iOS & Android) from one codebase |
| Backend | Node.js or Django (Python) | Fast development, large community support |
| Database | Firebase or PostgreSQL | Firebase for real-time updates; PostgreSQL for structured data |
| Maps | Google Maps API / Mapbox | Rich mapping features; free tier available |
| Version Control | GitHub | Collaboration, code review, and CI/CD integration |
| Project Management | Jira or Trello | Track tasks, sprints, and milestones |

**8. Project Timeline**

The following high-level timeline outlines the major phases of developments:

| Phase | Activities | Duration |
|---|---|---|
| Phase 1: Planning & Requirements | Finalize scope, stakeholder meetings, requirements gathering | Week 1 - 2 |
| Phase 2: Design & Prototyping | UI/UX wireframes, database schema, system architecture | Week 3 - 5 |
| Phase 3: Development (MVP) | Frontend & backend coding, API integration, map setup | Week 6 - 14 |
| Phase 4: Testing & QA | Unit tests, integration tests, user acceptance testing | Week 15 - 17 |
| Phase 5: Deployment & Launch | App store submission, server deployment, student rollout | Week 18 |

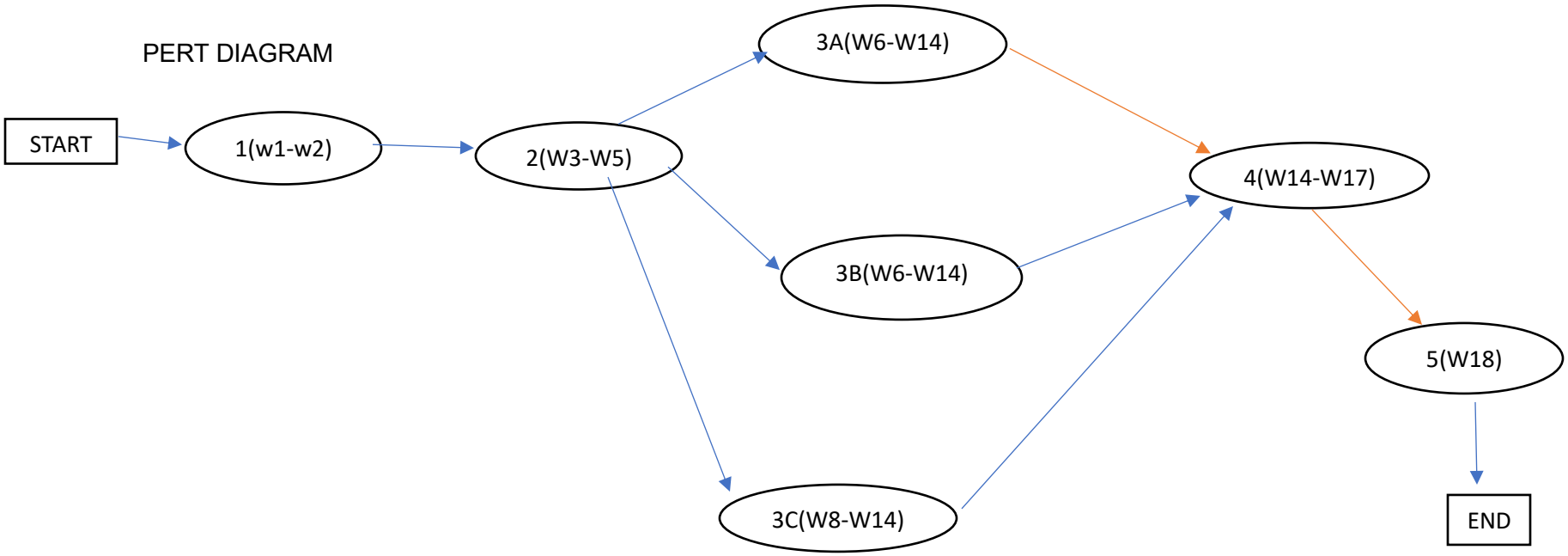
**GANTT CHART AND PERT DIAGRAM**

| Activity node | Activity | Depends on | Duration |
|---|---|---|---|
| **1** | Planning and requirements | none | 2 weeks |
| **2** | Design and prototyping | Node 1 | 3weeks |
| **3A** | Fronted development | Node 2 | 9 weeks |
| **3B** | Backend development | Node 2 | 9weeks |
| **3C** | Map and API Integration | Node 2 | 7weeks |
| **4** | Testing and QA | Nodes 3A, 3B, 3C | 3 weeks |

| **5** | Deployment and launch | Node 4 | 1 week |
|---|---|---|---|

| | weeks | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 |
| 1 | | | | | | | | | | | | | | | | | | |
| 2 | | | | | | | | | | | | | | | | | | |
| 3A | | | | | | | | | | | | | | | | | | |
| 3B | | | | | | | | | | | | | | | | | | |
| 3C | | | | | | | | | | | | | | | | | | |
| 4 | | | | | | | | | | | | | | | | | | |
| 5 | | | | | | | | | | | | | | | | | | |

PERT DIAGRAM



**9.Immediate Actions**

Following approval, the team will initiate the following:

- Host a kickoff meeting to align stakeholders.
- Define roles and establish the GitHub repository and management board.
- Validate requirements via stakeholder interviews and student surveys.
- Commence the design phase, including user flows and database schema.